② Calcule $\mathcal{L}^{-1}[F(s)]$, para:

a) $F(s) = \dfrac{e^{-s}}{s(s+1)}$

Sabemos que $f(t-a)H(t-a) \rightarrow \mathcal{L}\left[\dfrac{e^{-as}}{s}\right]$

$\mathcal{L}^{-1}\left[e^{-as}F(s)\right] \rightarrow H(t-a)f(t-a)$

Para $\dfrac{e^{-s}}{s(s+1)}$: $F(s) = \dfrac{1}{s(s+1)}$, $a=1$

propriedade:

$H(t-1)\mathcal{L}^{-1}\left[\underbrace{\dfrac{1}{s(s+1)}}_{*}\right](t-1) \rightarrow \mathcal{L}^{-1}\left[\dfrac{1}{s-a}\right] = e^{at}$

(*) resolvendo $\mathcal{L}^{-1}\left[\dfrac{1}{s(s+1)}\right]$. Usando fração parcial:

$\dfrac{1}{s(s+1)} = \dfrac{A}{s} + \dfrac{B}{s+1} \rightarrow \dfrac{1 = (s+1)A + Bs}{s(s+1)}$

com $s = -1 \rightarrow 1 = (-1+1)A + B(-1) \rightarrow B = -1$

achando A $\rightarrow s = 1 \rightarrow 1 = (1+1)A + B(1) \rightarrow 1 = 2A + B \rightarrow 1 = 2A - 1$

$\rightarrow \boxed{A = 1}$

$\dfrac{A}{s} + \dfrac{B}{s+1} \rightarrow \dfrac{1}{s} - \dfrac{1}{s+1}$

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$\mathcal{L}^{-1}\left[\frac{1}{s-a}\right] = e^{-at}$

$\mathcal{L}^{-1}\left[\frac{1}{s}\right] = H(t)$ $\quad$ $\mathcal{L}^{-1}\left[\frac{1}{s+1}\right] = e^{-t}$

voltando em $H(t-1)\mathcal{L}^{-1}\left[\frac{1}{s(s+1)}\right](t-1)$

$= H(t-1)\left(\mathcal{L}^{-1}\left[\frac{1}{s}\right] - \mathcal{L}^{-1}\left[\frac{1}{s+1}\right]\right)(t-1)$

$= H(t-1)\left(H(t) - e^{-t}\right)(t-1)$

$= H(t-1)\left(H(t-1) - e^{-(t-1)}\right)$

b) $F(s) = \frac{2s-1}{s^2(s+1)^3}$

Primeiro devemos tirar a fração parcial:

$$\frac{2s-1}{s^2(s+1)^3} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+1} + \frac{D}{(s+1)^2} + \frac{E}{(s+1)^3}$$

$$\frac{2s-1}{s^2(s+1)^3} = \frac{A[s(s+1)^3] + B(s+1)^3 + C[s^2(s+1)^2] + D[s^2(s+1)] + Es^2}{s^2(s+1)^3}$$

$s=0 \rightarrow 2(0)-1 = A(0) + B(1) + C(0) + D(0) + E(0) \rightarrow \boxed{B=-1}$

$s=1 \rightarrow 2(1)-1 = A[1(1+1)^3] + B(2)^3 + C[1^2(2)^2] + D[1^2(2)] + E(1^2)$

$1 = 8A + 8B + 4C + 2D + E$

$1 = 8A - 8 + 4C + 2D + E \rightarrow 9 = 8A + 4C + 2D + E$

$s=-1 \rightarrow 2(-1)-1 = A[-1(-1+1)^3] + B(-1+1)^3 + C[(-1)^2(-1+1)^2] + D[(-1)^2(-1+1)] + E(-1)^2$

$-3 = E \rightarrow \boxed{E=-3}$

$s=2 \rightarrow 2(2)-1 = A[2(2+1)^3] + B(2+1)^3 + C[(2)^2(2+1)^2] + D[(2)^2(2+1)] + E(2)^2$

$3 = 54A + 27B + 36C + 12D + 4E$

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de $s=1 \rightarrow 9 = 8A + 4C + 2D + E$

$9 = 8A + 4C + 2D - 3$

$12 = 8A + 4C + 2D \quad /\cdot 2$

$6 = 4A + 2C + D$, com $A=5$, $C=-5$,

$6 = 4(5) + 2(-5) - 4$

$6 = 20 - 10 - 4 \rightarrow 6 = 6$

$A=5,\ B=-1,\ C=-5,\ D=-4,\ E=-3$

$$\mathcal{L}^{-1}\left[\frac{5}{s} - \frac{1}{s^2} - \frac{5}{s+1} - \frac{4}{(s+1)^2} - \frac{3}{(s+1)^3}\right]$$

$$\mathcal{L}^{-1}\left[\frac{5}{s}\right] - \mathcal{L}^{-1}\left[\frac{1}{s^2}\right] - \mathcal{L}^{-1}\left[\frac{5}{s+1}\right] - \mathcal{L}^{-1}\left[\frac{4}{(s+1)^2}\right] - \mathcal{L}^{-1}\left[\frac{3}{(s+1)^3}\right]$$

① ② ③ ④ ⑤

① $\mathcal{L}^{-1}\left[\frac{5}{s}\right] = 5H(t)$

② $\mathcal{L}^{-1}\left[\frac{1}{s^2}\right] = t$

③ $\mathcal{L}^{-1}\left[\frac{5}{s+1}\right] = 5\mathcal{L}^{-1}\left[\frac{1}{s+1}\right] = 5e^{-t}$

④ $\mathcal{L}^{-1}\left[\frac{4}{(s+1)^2}\right] = \mathcal{L}^{-1}[F(s-a)] = e^{at} f(t) = e^{-t}\mathcal{L}^{-1}\left[\frac{4}{s^2}\right]$

$a=1$

$= \{e^{-t} \cdot 4t\}$

⑤ $\mathcal{L}^{-1}\left[\frac{3}{(s+1)^3}\right] = \mathcal{L}^{-1}[F(s-a)] = e^{at} f(t) = e^{-t}\mathcal{L}^{-1}\left[\frac{3}{s^3}\right]$

$a=-1$

$= \left\{e^{-t} \cdot \frac{3t^2}{2}\right\}$

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Sol.: $s_H(t) = t - 5e^{-t} - 4te^{-t} - \frac{3e^{-t}t^2}{2}$

c) $F(s) = \frac{s}{s^2+2s-3}$

$$\frac{s}{(s-1)(s+3)} = \frac{A(s+3)}{(s-1)} + \frac{B(s-1)}{(s+3)}$$

$s = 1 \rightarrow s = A(s+3) + B(s-1)$

$1 = A(1+3) + B(1-1)$

$4A = 1 \rightarrow A = \frac{1}{4}$

$s = -1 \rightarrow -1 = A(-1+3) + B(-1-1)$

$-1 = \frac{1}{4}(2) - 2B$

$-1 - \frac{1}{2} = -2B \rightarrow \frac{-2-1}{2} = -4B \rightarrow -3 = -4B$

$B = \frac{3}{4}$

$$\frac{1}{4(s-1)} + \frac{3}{4(s+3)}$$

$$\mathcal{L}^{-1}\left[\frac{1}{4(s-1)}\right] + \mathcal{L}^{-1}\left[\frac{3}{4(s+3)}\right]$$

$$\equiv \frac{1}{4}e^{t} + \frac{3}{4}e^{-3t}$$

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